

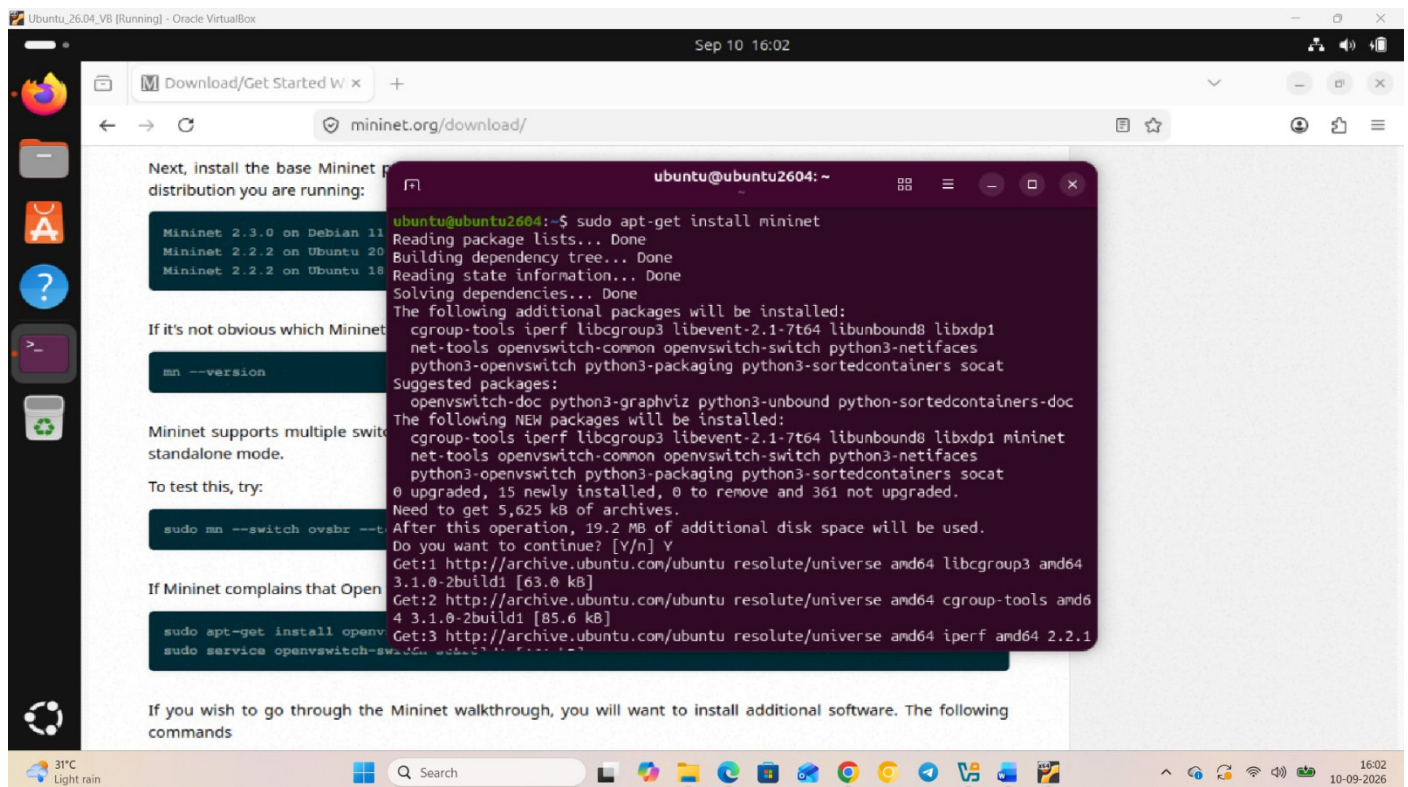
Assignment 3: Creation of a Simple Network Topology Using Open-Source Network Virtualization Tools

Objective To create and configure a simple network topology using an open-source network virtualization tool and verify communication between the virtual network nodes.

Task

- Ubuntu Linux Virtual Machine
- VirtualBox
- Mininet
- Create simple Network Topology
- ping and basic Linux networking commands
- After completion, stop the network.

Install MININET: <https://mininet.org/download/>



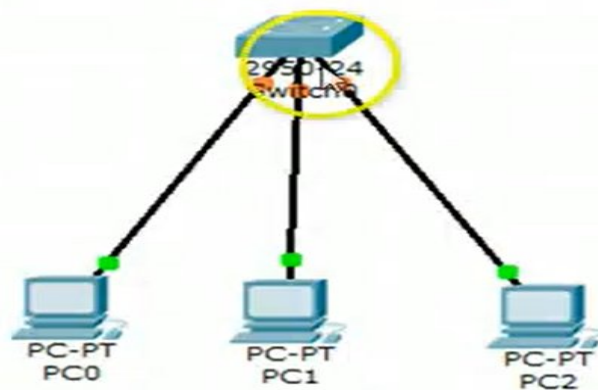
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Ubuntu_26.04_VB [Running] - Oracle VirtualBox
Sep 10 16:20
ubuntu@ubuntu2604: ~ -- sudo mn --topo single,3

-t WAIT, --twait=WAIT
    timed wait (s) for switches to connect
--cluster=server1,server2...
    run on multiple servers (experimental!)
--placement=block|random
    node placement for --cluster (experimental!)

ubuntu@ubuntu2604:~$ sudo mn --topo single,3
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller

*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> links
h1-eth0<->s1-eth1 (OK OK)
h2-eth0<->s1-eth2 (OK OK)
h3-eth0<->s1-eth3 (OK OK)
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet>
```

single,3



```
Ubuntu_26.04_V8 [Running] - Oracle VirtualBox
Sep 10 16:33
ubuntu@ubuntu2604: ~ — sudo mn --topo tree,4

h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> exit
*** Stopping 0 controllers

*** Stopping 3 links
...
*** Stopping 1 switches
s1
*** Stopping 3 hosts
h1 h2 h3
*** Done
completed in 1003.392 seconds
ubuntu@ubuntu2604:~$ sudo mn --topo tree,4
[sudo: authenticat] Password:
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16
*** Adding switches:
s1 s2 s3 s4 s5 s6 s7 s8 s9 s10 s11 s12 s13 s14 s15
*** Adding links:
(s1, s2) (s1, s9) (s2, s3) (s2, s6) (s3, s4) (s3, s5) (s4, h1) (s4, h2) (s5, h3) (s5, h4) (s6, s7) (s6, s8) (s7, h5) (s7, h6) (s8, h7) (s8, h8) (s9, s10) (s9, s13) (s10, s11) (s10, s12) (s11, h9) (s11, h10) (s12, h11) (s12, h12) (s13, s14) (s13, s15) (s14, h13) (s14, h14) (s15, h15) (s15, h16)
*** Configuring hosts
h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16
*** Starting controller

*** Starting 15 switches
s1 s2 s3 s4 s5 s6 s7 s8 s9 s10 s11 s12 s13 s14 s15 ...
*** Starting CLI:
mininet>
```

```
Ubuntu_26.04_V8 [Running] - Oracle VirtualBox
Sep 10 16:34
ubuntu@ubuntu2604: ~ — sudo mn --topo tree,4

*** Starting 15 switches
s1 s2 s3 s4 s5 s6 s7 s8 s9 s10 s11 s12 s13 s14 s15 ...
*** Starting CLI:
mininet> links
s1-eth1<->s2-eth3 (OK OK)
s1-eth2<->s9-eth3 (OK OK)
s2-eth1<->s3-eth3 (OK OK)
s2-eth2<->s6-eth3 (OK OK)
s3-eth1<->s4-eth3 (OK OK)
s3-eth2<->s5-eth3 (OK OK)
s4-eth1<->h1-eth0 (OK OK)
s4-eth2<->h2-eth0 (OK OK)
s5-eth1<->h3-eth0 (OK OK)
s5-eth2<->h4-eth0 (OK OK)
s6-eth1<->s7-eth3 (OK OK)
s6-eth2<->s8-eth3 (OK OK)
s7-eth1<->h5-eth0 (OK OK)
s7-eth2<->h6-eth0 (OK OK)
s8-eth1<->h7-eth0 (OK OK)
s8-eth2<->h8-eth0 (OK OK)
s9-eth1<->s10-eth3 (OK OK)
s9-eth2<->s13-eth3 (OK OK)
s10-eth1<->s11-eth3 (OK OK)
s10-eth2<->s12-eth3 (OK OK)
s11-eth1<->h9-eth0 (OK OK)
s11-eth2<->h10-eth0 (OK OK)
s12-eth1<->h11-eth0 (OK OK)
s12-eth2<->h12-eth0 (OK OK)
s13-eth1<->s14-eth3 (OK OK)
s13-eth2<->s15-eth3 (OK OK)
s14-eth1<->h13-eth0 (OK OK)
s14-eth2<->h14-eth0 (OK OK)
s15-eth1<->h15-eth0 (OK OK)
s15-eth2<->h16-eth0 (OK OK)
mininet>
```

Ubuntu_26.04_V8 [Running] - Oracle VirtualBox

Sep 10 16:35

ubuntu@ubuntu2604: ~ — sudo mn --topo tree,4

```
s8-eth2<->h8-eth0 (OK OK)
s9-eth1<->s10-eth3 (OK OK)
s9-eth2<->s13-eth3 (OK OK)
s10-eth1<->s11-eth3 (OK OK)
s10-eth2<->s12-eth3 (OK OK)
s11-eth1<->h9-eth0 (OK OK)
s11-eth2<->h10-eth0 (OK OK)
s12-eth1<->h11-eth0 (OK OK)
s12-eth2<->h12-eth0 (OK OK)
s13-eth1<->s14-eth3 (OK OK)
s13-eth2<->s15-eth3 (OK OK)
s14-eth1<->h13-eth0 (OK OK)
s14-eth2<->h14-eth0 (OK OK)
s15-eth1<->h15-eth0 (OK OK)
s15-eth2<->h16-eth0 (OK OK)
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16
h2 -> h1 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16
h3 -> h1 h2 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16
h4 -> h1 h2 h3 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16
h5 -> h1 h2 h3 h4 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16
h6 -> h1 h2 h3 h4 h5 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16
h7 -> h1 h2 h3 h4 h5 h6 h8 h9 h10 h11 h12 h13 h14 h15 h16
h8 -> h1 h2 h3 h4 h5 h6 h7 h9 h10 h11 h12 h13 h14 h15 h16
h9 -> h1 h2 h3 h4 h5 h6 h7 h8 h10 h11 h12 h13 h14 h15 h16
h10 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h11 h12 h13 h14 h15 h16
h11 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h12 h13 h14 h15 h16
h12 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h13 h14 h15 h16
h13 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h14 h15 h16
h14 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h15 h16
h15 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h16
h16 -> h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15
*** Results: 0% dropped (240/240 received)
mininet>
```

31°C
Light rain

Search

16:35
10-09-2026

tree,3

